Name _____ _____ Baggett/Fall2026/207/Quiz4

first name last name

Score

No Calculators. / 20

Questions 1 and 2. Circle **True** or **False**. Then, in **one sentence**, say why — if the statement is false, the fastest way is usually to give an example that breaks it. **1 point for the circle, 1 for the reason.**

1. (**2** pts) $\dfrac{d}{dx}\left[\dfrac{f(x)}{g(x)}\right] = \dfrac{f'(x)}{g'(x)}$.

True **False**

Why? _____ /2

2. (**2** pts) If $f(x) = \sin x$, then $f^{(19)}(x) = \cos x$.

True **False**

Why? _____ /2

3. (**5** pts) Let $f(x) = x\cos x$.

(a) Find $f'(x)$.

(b) Find the equation of the **tangent line** to $y = f(x)$ at $x = \pi$.

_____ /5

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4. (**5** pts) Differentiate $f(x) = \dfrac{\sin x}{x^2 + 1}$.

/5

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5. (**3** pts) Differentiate $y = x^3 + e^x \cos x + 2x \sin x$.

/3

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6. (**3** pts) For each function below, name the **rule you would use first**. **Do not differentiate anything.**

(a) $y = x^2 \cos x$ _____

(b) $y = \dfrac{x^2}{3}$ _____

(c) $y = \dfrac{\cos x}{x^2}$ _____

/3

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